



Airbus Product Standard

EP-Prepreg Glass fibre fabric preimpregnated with epoxy resin

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1 Scope

This standard specifies the required characteristics of prepreg made from E-glass fibre fabric preimpregnated with epoxy or modified resin for structural and non-structural applications.

2 Normative References

This Airbus Standard incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any these publications apply to this Airbus Standard only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

ABS5940	Airbus Standard – E glass fibre fabric preimpregnated with epoxy or modified resin. Out of autoclave.
AIMS05-02-000 Part 1	Technical Specification – Glass fibre reinforced epoxy prepreg – General requirements
AIMS05-02-002	Material Specification – Glass fibre reinforced thermosetting prepreg – Glass fabric US Style 120 / epoxy resin, 180 °C curing class (105 g/m ² fibre areal weight)
AIMS05-02-003	Material Specification – Glass fibre reinforced thermosetting prepreg Glass fabric US Style 1581/7581/7781 / epoxy resin, 180 °C curing class (300 g/m ² fibre areal weight)
AIMS05-02-004	Material Specification – Glass fibre reinforced thermosetting prepreg Glass fabric / epoxy resin, 120 °C cure temperature (105 g/m ² fibre area mass). Non-structural application
AIMS05-02-005	Material Specification – Glass fibre reinforced thermosetting prepreg – Glass fabric US Style 1543 / epoxy resin, 180 °C curing class (296 g/m ² fibre areal weight)
AIMS05-02-006	Material Specification – Glass fibre reinforced thermosetting prepreg Glass fabric US Style 1080 / epoxy resin, 180 °C curing class (48 g/m ² fibre areal weight)
AIMS05-02-007	Material Specification – Glass fiber reinforced thermosetting prepreg. Glass fabric / Epoxy resin. 120 °C Cure temperature (104 g/m ² fiber area mass). Structural application
DAN1164	Material Specification - EP-Prepreg: woven fabric of E-glass filament yarn 8.4544.60 preimpregnated with epoxy resin (EP-resin) – 125 °C and 180 °C curing systems, begin of cure 60 °C
DAN1227	Material Specification - EP-Prepreg: woven fabric of E-glass filament yarn 8.4544.60 preimpregnated with epoxy resin (EP-resin) – 125 °C system
DAN1242	Material Specification - EP-Prepreg: woven fabric of E-glass filament yarn 8.4565.60, 8.4567.60, 8.4568.60 preimpregnated with epoxy resin (EP-resin) – 125 °C system
DAN1243	Material Specification - EP-Prepreg: woven fabric of E-glass filament yarn 8.4546.60 preimpregnated with epoxy resin (EP-resin) – 125 °C system
DAN1252	Material Specification - EP-Prepreg: preimpregnated E-glass, insulation layer
DAN463	Technical Specification Preimpregnated filament fabric from E-glass or aramid fiber and EP-resin for structural applications
DIN65090	Aerospace; textile glass; preimpregnated filament glass cloth from E-glass (Prepreg) – Technical Specification



3 Requirements

3.1 Configuration, dimensions, tolerances and mass

The configuration, dimensions, tolerances and mass shall be in accordance with Table 1.

Table 1: Dimensions, tolerances and mass

Dimensions in mm

Delivery width ^{c)} ± 1,5%		Delivery length ^{a)} ± 10%	Inner diameter of cores min.	Mass per roll kg max.
6,35 ^{d)}	trimmed	≤ 1 500 000	75	5,5
12,7 ^{e)}		≤ 800 000		
150 ^{b)}	trimmed	≤ 100 000	254	50
300 ^{b)}			75	
460				
700				
720				
965				
1000				
1000	trimmed	≤ 50 000		
1070				
1100	untrimmed	≤ 100 000		
1170	trimmed			
1240				
1200	untrimmed			
1270				
1270	trimmed			
a) A roll may consist of 3 separate lengths at the most with a minimum length of 15 m per piece. The individual lengths must be marked at the ends. b) Width tolerance ^{+0/-0,75} mm, ATL grade c) As per AIMS05-02-000 Part 1 §4.1.11.3 d) Width tolerance ± 0,2 mm, slit fabric e) Width tolerance ± 0,3 mm, slit fabric				



3.2 Material and surface treatment

The material shall be in accordance with Table 2, Table 3 and Table 4

Table 2: Material 125°C/180°C cure

Material code	Material	Nominal cured ply thickness [mm]	Mass per unit area [g/m ²]	Resin content [wt%]
A50 a) b)	DAN1164-50	0,10	max. 250	50
a) 60 °C start cure b) Not for new design				

Table 3: Material 125°C cure

Material code	Material	Nominal cured ply thickness [mm]	Mass per unit area [g/m ²]	Resin content [wt%]
B37	DAN1227-37	0,10	max. 190	37
B53	DAN1227-53		max. 240	53
B55	DAN1227-55		max. 285	55
C37	DAN1242-37	0,25	max. 510	37
C40	DAN1242-40		max. 520	40
C50	DAN1242-50		max. 650	50
D37	DAN1243-37		max. 500	37
E37	DAN1252-37	0,10	max. 180	37
J39	AIMS05-02-004		max. 205	42
L38	AIMS05-02-007A	0,090	max. 185	38
L45	AIMS05-02-007B	0,104	max. 203	45
L50	AIMS05-02-007C	0,120	max. 230	50

AIRBUS**Table 4: Material 180°C cure**

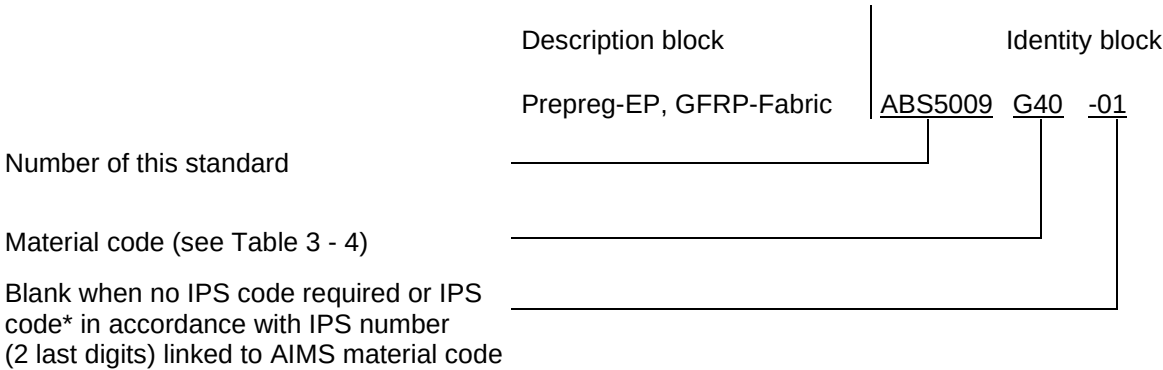
Material code	Material	Nominal cured ply thickness [mm]	Mass per unit area [g/m ²]	Resin content [wt%]
J56 ^{c)}	AIMS05-02-006A	0,066 ^{d)}	max.118	56
E40	DAN1252-40	0,10	max. 190	40
E50	DAN1252-50		max. 220	50
F38 ^{a)}	AIMS05-02-002A		max. 210	40 - 45
F40 ^{b)}	AIMS05-02-003C	0,25	max.505	37
G38 ^{b)}	AIMS05-02-003A/B		max. 518	37 - 40
G40	AIMS05-02-005A		max. 505	37
H40 ^{e)}	AIMS05-02-002A	0,10	max. 183	40
H42	AIMS05-02-002A		max. 190	42
H45	AIMS05-02-002A	0,11	max. 210	45
K40 ^{c) e) f)}	AIMS05-02-002A	0,10	max. 190	40
^{a)} Not for new design ^{b)} AIMS05-02-003A/B (G38) for non-structural application, AIMS05-02-003C (F40) for structural application ^{c)} See relevant IPS for specific galvanic corrosion protection usage ^{d)} CPT corrected from 0,05 to 0,066. See Table 5 for codes of equivalence ^{e)} The H40 code refers to all AIMS05-02-002 materials with 40% resin content with the exception of IPS05-02-002-05, for which the K40 code is used. ^{f)} Not for new design refer to ABS5940, see also Table 5.				



4 Designation

4.1 Current designation

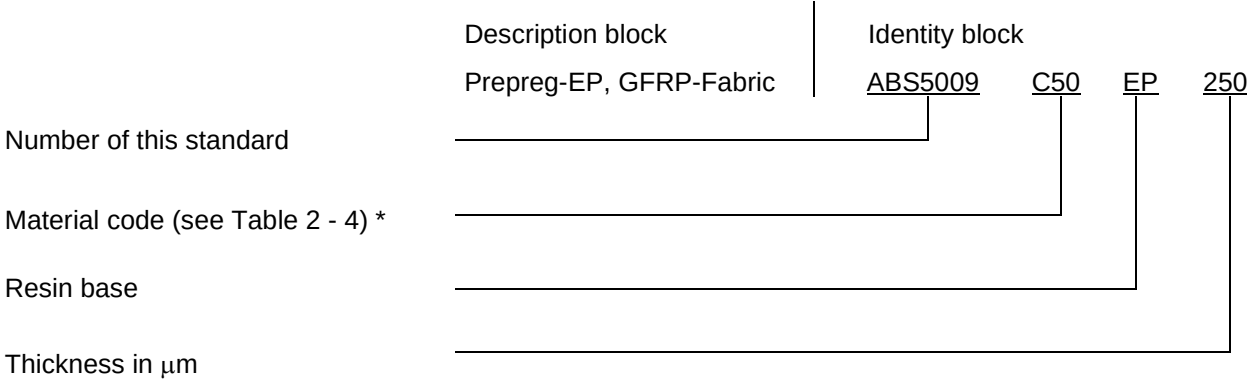
This type of Standard shall be designated according to the philosophy of the following example:



***Note:** Designation with IPS code when required by the design

4.2 Previous designation

The previous designation used for this Standard was designated according to the philosophy of the following example:



***Note:** IPS-identification is not possible, refer to relevant drawing for any applicable IPS designation.



Table 5: Codes of Equivalence

Airbus drawing codes	
Previous designation (Column 1)	Current applicable designation (Column 2)
ABS5009J56EP050	ABS5009J56EP066
ABS5009F38-02	ABS5009H40-02
ABS5009F38-04	ABS5009H45-04
ABS5009K40EP100	ABS5940A0000
Column 2 is the current applicable designation for drawings. In cases where the previous drawing designations has been used, column 1 provides information of applicable cross references	

5 Technical Specification

Technical Specification AIMS05-02-000 Part 1, DAN463 and DIN65090 as applicable.



Record of Revisions

Issue	Clause modified	Description of Modification
1 10.1998		New Standard
2 06.2000		Material codes added: E50
3 12.2005	3.2 4.2	Material codes added: F38-01, F38-02, F38-04, F40, G38, G38-02, G38-04, H40, H45 Clause added: "specific designation for Airbus-France"
4 11.2007	2 3.2	Specification AIMS05-02-004 added Material codes added: J39, H42
5 01.2010	2 3.2	Specification AIMS05-02-006 added Material codes added: J56
6 05.2010	Table 1 Table 2	Addition of new material widths Update to J56 material nominal cured ply thickness [mm]
7 08.2010	Table 2 4	Note 6) added for change in CPT from Issue 6 Codes of Equivalence table added
8 09.2011	Table 2	Material codes added: J40
9 11.2011	Table 2	Note 5) added to J40 code for galvanic corrosion requirements Note 7) added for resin content
10 01.2012	Table 1	Tolerances modified New width added: 150, 700 and 1070
11 03.2012	Table 2	Code 40 added New code K54 added Note 7) removed Note 8) added Removal of J40 code, details added to K40 code
12 12.2014	2 Table 3 Table 4 4.2 Table 5	Reference to ABS5940 and AIMS05-02-007 added Codes added: L38, L45, L50 Notes 9), 10) and 11) added. Note 3 removed. Note 8 changed Chapter removed. ABS full code designation added Codes of equivalence updated
13 08.2015	Table 4 Table 5	Editorial revision of footnotes, removal of K54 code Codes of equivalence updated



Issue	Clause modified	Description of Modification
14 03.2019	Table 3	Updated to latest template and reformatted as required Resin content material code J39 updated.
15 04.2020	3.1 3.2 4.1 4.2	Table 1: new widths added (slit fabric) including footnotes for tolerances Table 4: Material grades for all materials added, CPT information for H45 added to relevant cell, footnotes amended Chapter changed to “Current designation” Chapter changed to “Previous designation”; Table 5 revised
16 03.2025	3.2	Table 3: Under row J39, resin content was changed to 42%